

Launch-Moment Bounds for a Wider Conditional MQM Thermal Capture Window

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Abstract—The fixed vacuum launch of the MQM boundary latch has conserved generator moments much smaller than its full spectral norm. In the source rotating frame, $\|H\psi_0\| = 1$ and $\|H^2\psi_0\| = \sqrt{2}$. These identities imply a unit live-amplitude Lipschitz bound and a population-curvature bound. They reduce the cold variation from $49w/1024 + 49w^2$ to $7w/1024 + w^2$. With the inherited response jet, Gaussian remainder and uniformly extended continuum value comparisons, exact arithmetic certifies retained thermal error below $127/2^{23} < 2^{-16}$ throughout $[16 - 2^{-11}, 16 + 2^{-11}]$ at unchanged $\beta = 5/16$. This widens the preceding halfwidth eightfold. No numerical continuum-derivative transfer, permanent retention, arbitrary-input diamond bound or physical implementation is claimed.

Index Terms—Quantum control, vacuum launch, spectral moments, finite retention.

I. CONTRACT AND PROVENANCE

Keep $\Omega = 64$, $B = 16$, $\kappa = 8$, $\beta = 5/16$, $T = 16$, $h = g = 1$, and $\lambda^2 = 4\sqrt{2}$. The initial state is $\psi_0 = |p0, \text{vac}\rangle$. Let P project onto the four live coordinates in the conserved one-excitation sector. The port joins the fourth live coordinate to the band reservoir; the first two native edges have unit weights.

The generator is the centered rotating-frame vacuum generator, not an absolute-energy Hamiltonian. Scalar shifts preserve $q(t) = \|P\psi(t)\|^2$ but change uncentered moments; the identities below use the declared frame. Mathematical modes are not physical qubits. Spectators remain identities.

The parent release [1] certifies conditional halfwidth 2^{-14} , its numerical continuum-derivative transfer remains open. This release uses an analytic vacuum identity and uniform response-value comparisons. Window selection is post-exposure engineering certification, not independent prediction. The enlarged list 2^{-k} , $k = 8, \dots, 24$, was recorded before this release's evaluation.

II. EXACT LAUNCH MOMENTS

In every admitted finite reference and its flat-band continuum counterpart,

$$H\psi_0 = |p1, \text{vac}\rangle, \quad (1)$$

$$H^2\psi_0 = |p0, \text{vac}\rangle + |p2, \text{vac}\rangle. \quad (2)$$

No bath coefficient or λ enters these identities. Orthonormality gives

$$\langle H \rangle_0 = 0, \quad \langle H^2 \rangle_0 = 1, \quad \langle H^4 \rangle_0 = 2. \quad (3)$$

Unitary evolution commutes with H , so $\|H\psi(t)\| = 1$ and $\|H^2\psi(t)\| = \sqrt{2}$. The finite-band one-excitation generator is

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bounded; these differentiations are well-defined. The moments apply to the named launch, not arbitrary input or unrestricted thermal Fock space.

Theorem 1. *The live amplitude is 1-Lipschitz, and*

$$|q'(t)| \leq 2\sqrt{q(t)}, \quad |q''(t)| \leq 2 + 2\sqrt{2q(t)}. \quad (4)$$

Proof. For $\psi = e^{-iHt}\psi_0$, $\|(P\psi)'\| \leq \|H\psi\| = 1$. Integrating and using the reverse triangle inequality proves the amplitude statement, including its zeros. Differentiation gives

$$q' = 2\Re\langle P\psi, -iPH\psi \rangle, \quad (5)$$

$$q'' = 2\|PH\psi\|^2 - 2\Re\langle P\psi, PH^2\psi \rangle. \quad (6)$$

Cauchy–Schwarz and the conserved moments imply both bounds. $\square$

No novelty comparison with general quantum speed-limit results has been performed.

III. LOCAL POPULATION ALLOWANCE

The inherited point certificate [1], [2] gives

$$q(T) < E_0 = 3/2^{18} + 2^{-128} < (7/2048)^2. \quad (7)$$

Set $a_0 = 7/2048$. On $|t - T| \leq w$,

$$\sqrt{q(t)} < a_0 + w, \quad (8)$$

$$q(t) < E_0 + d_q(w), \quad (9)$$

$$d_q(w) = 2a_0w + w^2 = 7w/1024 + w^2. \quad (10)$$

The previous port bound used $G < 7$ and $49w/1024 + 49w^2$. The launch moment improves the linear charge sevenfold and quadratic charge forty-ninefold without modifying the dynamics.

A separate Taylor route uses $|q'(T)| < 2a_0$, $\sqrt{2} < 3/2$, and the interval amplitude bound:

$$d_{q,\text{curv}}(w) = 2a_0w + [1 + \frac{3}{2}(a_0 + w)]w^2. \quad (11)$$

It is slightly more conservative and gives the same passing dyadic candidate.

IV. UNIFORM COMPARISON AND THERMAL LEDGER

The inherited finite-response certificate [1] remains

$$\frac{1}{2}\|\Delta_{2,M}(t) - \Delta_{2,M}(T)\|_1 \leq 2^{-20}w + w^2/8, \quad M = 192, \quad (12)$$

with point response below $3/2^{26}$. Extend the uniform comparison horizon to $U_* = 4097/256 = 16 + 2^{-8}$, covering the declared list. The centered embedding [3] uses

$$R = \frac{x^{192}}{192! [1 - x/193]}, \quad x = 4U_*. \quad (13)$$

Exact arithmetic verifies $336U_*^3 \bar{V} R < 2^{-30}$, where $\bar{V} = 64/31457265$. For the distinct stationary covariance, put $r = 2^{-21}$ and $y = 8U_* + 25/16$:

$$\eta = \frac{4(128/3)r}{1-r} \frac{y^{384}}{384! [1 - y/385]} \quad (14)$$

$$+ \frac{2(128/3)r^9}{1-r}, \quad 2U_*^2 \eta < 2^{-57}. \quad (15)$$

These compare values, not derivatives. The Gaussian remainder is $R_{\geq 4} = x_U^2/[2(1 - x_U)]$, with

$$x_U = x_{16} + 6\bar{V}(Tw + w^2/2), \quad (16)$$

$$x_{16} = 12433696189517/58512905602490430. \quad (17)$$

Theorem 2. *Under the inherited fixed-launch continuum, Gaussian and arithmetic assumptions, retained thermal error is below $127/2^{23} < 2^{-16}$ throughout 16 ± 2^{-11} model time units.*

Proof. At $w = 2^{-11}$, exact rational comparison establishes

$$E_\beta(t) < E_0 + d_q(w) + 3/2^{26} + 2^{-20}w + w^2/8 \quad (18)$$

$$+ 2^{-30} + 2^{-57} + R_{\geq 4}(T + w) \quad (19)$$

$$< 127/2^{23} < 2^{-16}. \quad (20)$$

The target also passes with $d_{q,\text{curv}}$. The comparison applies since $T + w \leq U_*$. Larger declared candidates fail this sufficient majorant, not the physical task. $\square$

The halfwidth is eight times [1] and thirty-two times [2], relative halfwidth 2^{-15} and full width 2^{-10} in model time. No pulse-off or device-time calibration is added.

V. AUDIT AND NEXT GATE

Dense spectral evolution and independent exponential action at $M = 4, 8, 192$ agree at nine declared times within 1.68×10^{-13} in vector norm. Commutator/vector population derivatives and curvature agree within 2.23×10^{-16} and 1.56×10^{-15} . Six moment identities pass; a separate integer audit checks seventeen decisions and the display ceiling. These are same-agent separate routes, not external review or continuum validation.

The finite $M = 192$ slope diagnostic near T is about -9.01×10^{-6} , much smaller than the generic certified ceiling. The next gate is to certify propagation of $H\psi_0 = |p1, \text{vac}\rangle$ and $H^2\psi_0$ using explicit continuum propagator comparison for each launch. Close values for the original launch alone do not close this bridge.

External review, permanent retention, bath refresh, physical two-local implementation and arbitrary-input diamond accuracy remain OPEN. Capacity allocation remains separate. Physical promotion remains zero.

REFERENCES

- [1] B. Mayes, *Retained-Response Derivative Bounds and a Wider Conditional MQM Thermal Capture Window*, v0.1, 2026.
- [2] B. Mayes, *A Finite Retention and Timing Window for the Conditional MQM Thermal Capture Certificate*, v0.1, 2026.
- [3] B. Mayes, *Time-Centered Gaussian Quadrature Embedding for the MQM Vacuum-Transfer Certificate*, v0.1, 2026.